# NEET 2026 (SAMPLE PAPER)

1. How many moles of magnesium phosphate will contain 0.2 mole of Oxygen atom?
   A. 0.02
   B. $3.125 \times 10^{-2}$
   C. $1.25 \times 10^{-2}$
   D. $2.5 \times 10^{-2}$

2. If 0.5 0 more of Barium Chloride is mixed with 0.20 mole of sodium phosphate, the maximum number of moles of Barium phosphate that can be format is
   A. 0.10
   B. 0.20
   C. 0.30
   D. 0.40

3. In Bohr series of lines of hydrogen spectrum, the third line from the red end corresponds to which one of the following inter-orbit jumps of the electron for Bohr orbits in an atom of Hydrogen?
   A. $3 \longrightarrow 2$
   B. $4 \longrightarrow 2$
   C. $5 \longrightarrow 2$
   D. $2 \longrightarrow 5$

4. Which of the following sets of quantum numbers represents the highest energy of an atom?
   A. n=3, l=2, m=+1, s=+1/2
   B. n=4, l=0, m=0, s=+1/2
   C. n=3,l=0, m=0, s=+1/2
   D. n=3, l=1, m=+1, s=+1/2

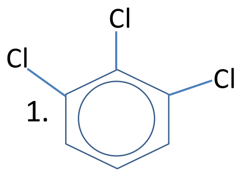
5. The electron affinity values (in Kj $mol^{-1}$) of three halogens X, Y and Z are respectively - 349 ,- 333 and -325.Then X, Y and Z respectively are
   A. $F_2$,$Cl_2$ and $Br_2$
   B. $Cl_2$,$F_2$ and $Br_2$
   C. $Cl_2$, $Br_2$ and $F_2$
   D. $Br_2$, $Cl_2$ and $F_2$

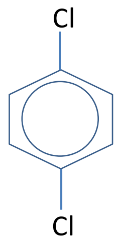
6. The first four ionization energy values of an element are 190,578 , 872 and 5960 kcal. The number of valence electrons in the element is
   A. 1
   B. 2
   C. 3
   D. 4

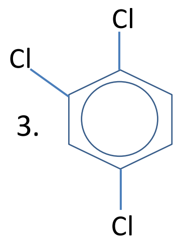
7. Which of the following statements is not correct for $\sigma$ and $\pi$ bonds which is formed between two carbon atoms?

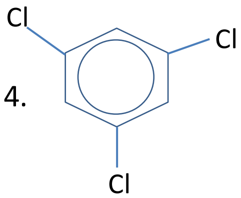
(A) Free rotation of atoms about $\sigma$ bond is allowed but not in case of $\pi$ bond.
(B) $\sigma$ bond determines the direction between carbon atoms but $\pi$ bond has no primary effect in this regard.
(C) $\sigma$ bond is stronger than a $\pi$ bond.
(D) Bond energies of $\sigma$ bond and $\pi$ bond are of the order of 264 kJ/mol and 347 kJ/mol, respectively.

8. Arrange the following compounds in the order of decreasing dipole moment

1. 


2. 


3. 


4. 


A. 1>3>2=4
B. 3>1>2=4
C. 1>2>3>4
D. 4>3>2>1

9. In which one of the following pairs, molecules/ions have similar shape?

(A) $CCl_4$, and $PtCl_4$,
(B) $NH_3$, and $BF_3$,
(C) $BF_3$, and t-butyl carbonium ion
(D) $CO_2$, and $H_2O$

10. The internal energy change when a system goes from state A to B is 40 kJ/mol. If the system goes from A to B by a reversible path and returns to state A by an irreversible path what would be the net change in internal energy?

(A) 40 kJ
(B) > 40kJ
(C) <40 kJ
(D) zero

11. From the colligative properties of solution which one is the best method for the determination of molar mass of proteins and polymers:

(A) osmotic pressure
(B) lowering in vapour pressure
(C) lowering in freezing point
(D) elevation in boiling point

12. During depression of freezing point in a solution the following are in equilibrium:
(A) liquid solvent, solid solvent
(B) liquid solvent, solid solute
(C) liquid solute, solid solute
(D) liquid solute, solid solvent

13. Which of the following liquid pairs shows a positive deviation from Raoult's law?
(A) Acetone-chloroform
(B) Benzene methanol
(C) Water-nitric acid
(D) Water-hydrochloric acid

14. For the reaction
$N_2(g)+3H_2(g) \rightleftarrows 2NH_3(g)$ , $\Delta H = - 93.6$ kJmo$^{-1}$ , the concentration of $H_2$ at equilibrium can be increased by

I. lowering the temperature
II. increasing the volume of the system
III. adding N2 at constant volume
IV. adding $NH_3$ at constant volume

A. II and IV are correct
B. Only II is correct
C. I, II and III are correct
D. III and IV are correct

15. When 10 mL of 0.1 M acetic acid (pKa = 5.0) is titrated against 10 mL of 0.1 M ammonia solution (pKb = 5.0) the equivalence point occurs at pH:

A. 5.0
B. 6.0
C. 9.0
D. 7.0

16. The solubility product of AgI at 25°C is $1.0 \times 10^{-16}$. The solubility of AgI in $10^{-4}$ N solution of KI at 25°C is approximately:
A. $1.0 \times 10^{-6}$
B. $1.0 \times 10^{-3}$
C. $1.0 \times 10^{-12}$
D. $1.0 \times 10^{-16}$

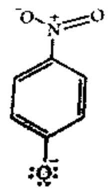
17. Given below are two statements:
Statement-I: Nitrous acid can act as oxidizing as well as reducing agent.
Statement-II: Oxidation number of nitrogen in nitrous acid is + 3, which is in between the lowest and highest possible oxidation numbers of Nitrogen.

A. Both statement-I and statement-II are correct
B. Both statement-I and statement-II are incorrect
C. Statement-I is correct but statement-II is incorrect
D. Statement-II is correct but statement-I is incorrect

18. Among the following substituted silanes, the one which will give rise to cross linked silicone polymer on hydrolysis is

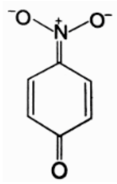
A. $R_4Si$
B. $RSiCl_3$
C. $R_2SiCl_2$
D. $R_3SiCl$

19. A structure which is not possible for p- nitrophenoxide ion is

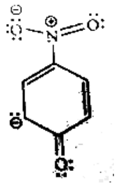


A.

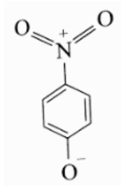
B.



C.



D.



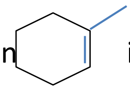
20.Match the List-I with List-II

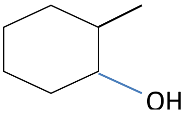
| | List-I | | List-II |
|---|---|---|---|
| **I** | **Electrophile** | **a** | **$:CCl_2$** |
| **II** | **Carbene** | **b** | **$SO_3$** |
| **III** | **Ambident nucleophile** | **c** | **$.CH_3$** |
| **IV** | **Free radical** | **d** | **$NO_2^-$** |

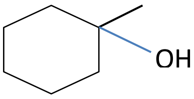
Choose the correct answer from the options given below:

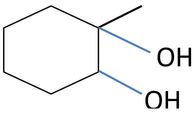
A. I-a , II-b , III- c, IV-d
B. I-a , II-c , III- b, IV-d
C. I-b , II-a , III- d, IV-c
D. I-d , II-b , III- c, IV-a

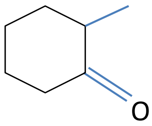
21. A compound with molecular formula $C_7H_{16}$ exhibits optical isomerism. The compound is
(A) 2,5 dimethylhexane
(B) 2-2-dimethylpentane
(C) 3-methylpentane
(D) 2,3-dimethylpentane

22. When  is treated with diborane in dry ether medium followed by oxidation with alkaline hydrogen peroxide, the major product formed will be

A. 


B. 


C. 


D. 


23. $CH3\text{-}C{\equiv}C\text{-}CH3 \xrightarrow[\text{Quinoline}]{H_2/\text{Pd-BaSO}_4} A \xrightarrow[CCl_4]{Br_2} B$ , The product (B) is

(A) meso-2,3-dibromo butane
(B) racemic-2, 3- dibromo butane
(C) 2,2,3,3-tetra bromo butane
(D) 1,4-dibromo butane

24. For the reduction of $NO_3^-$, ion in an aqueous solution, $E^0$ is +0.96 V. Values of $E^0$ for some metal ions are given below.

$V^{2+}(aq) + 2e^- \longrightarrow V \quad E^o = -1.19V$
$Fe^{3+}(aq) + 3e^- \longrightarrow Fe \quad E^o = -0.04\ V$
$Au^{3+}(aq) + 3e^- \longrightarrow Au \quad E^o = +1.40\ V$
$Hg^{2+}(aq) + 2e^- \longrightarrow Hg \quad E^o = +0.86\ V$

The metals that is/are oxidized by $NO_3^-$ in aqueous solution is/are

A. V,Hg,Au
B. V,Fe,Hg
C. Fe,Au,Hg
D. Only Au

25. The temperature dependence of rate constant (k) of a chemical reaction is written in terms of Arrhenius equation, $k = A.e^{-EA/RT}$, Activation energy ($E_A$) of the reaction can be calculated by ploting

A. log k vs $T^{-1}$
B. log k vs $\frac{1}{\log T}$
C. k vs T
D. k vs $\frac{1}{\log T}$

26. The decomposition of cyclopropane, was observed at 500$^0$ C and its concentration was monitored as a function of time. The data set is given below. What is the order of the reaction with respect to cyclopropane?

| Time (hour) | M [Cyclopropane], M |
|---|---|
| 0 | $1.00 \times 10^{-2}$ |
| 2 | $1.38 \times 10^{-3}$ |
| 4 | $1.91 \times 10^{-4}$ |
| 6 | $2.63 \times 10^{-5}$ |

A. 1
B. 2
C. 3
D. 0

27. A mechanism for a naturally occurring reaction that destroys ozone is

Step 1: $O_3 + HO(g) \longrightarrow HO_2(g) + O_2(g)$
Step II: $HO_2(g) + O(g) \longrightarrow HO(g) + O_2(g)$

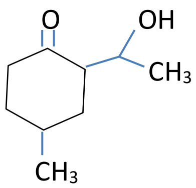
A. $HO_2$ ,heterogeneous
B. $HO_2$ ,homogeneous
C. HO ,heterogeneous
D. HO ,homogeneous

28. Which of the following statements is wrong?
A. Single N-N bond is stronger than the single P-P bond.
B. $PH_3$ can act as a ligand in the formation of coordination compound with transition elements.
C. $NO_2$ is paramagnetic in nature.
D. Covalency of nitrogen in $N_2O_5$ is four.

29. The radius of $La^{3+}$ (atomic number 57) is 1.06A. Which one of the following given values will be closest to the radius of $Lu^{3+}$ (atomic number = 71)?
A. 0.85 A$^0$
B. 1.06 A$^0$
C. 1.40 A$^0$
D. 1.60 A$^0$

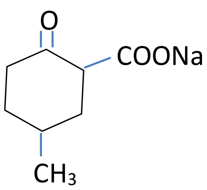
30. Which one is the most likely structure of $CrCl_3,6H_20$, if 1/3rd of total chlorine of the compound is precipitated by adding $AgNO_3$, to its aqueous solution?

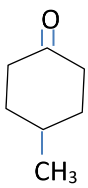
A. $[Cr(H_2O)_3Cl_3].(H_2O)_3$
B. $CrCl_3,6H_20$
C. $[Cr(H_2O)_5Cl].Cl_2(H_2O)$
D. $[Cr(H_2O)_4Cl_2].Cl(H_2O)_2$

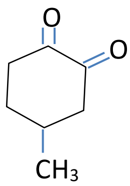
31. 

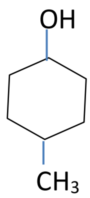
$\xrightarrow[\Delta]{I_2/NaOH} A \xrightarrow{H^+} B \xrightarrow{\Delta} C$

The final product 'C' is

A. 

B. 

C. 

D. 

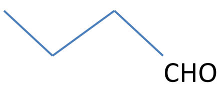

32. Which of the following is the correct order of dipole moment of methyl halides?

A. $CH_3F>CH_3Cl > CH_3Br > CH_3I$
B. $CH_3Cl>CH_3F > CH_3Br > CH_3I$
C. $CH_3F>CH_3Cl > CH_3I > CH_3Br$
D. $CH_3I>CH_3Cl > CH_3Br > CH_3F$

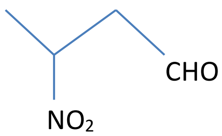
33. The major product formed when 2-Fluoropentane is treated with $C_2H_5O^-$/ $C_2H_5OH$ is

A. pent-1-ene
B. cis pent-2-ene
C. trans pent-2-ene
D. pentan-2-ol

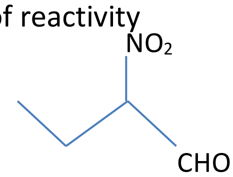
34. Arrange the following compounds in the increasing order of reactivity



I

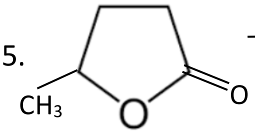


II



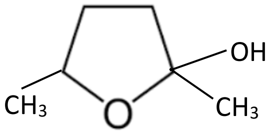
III

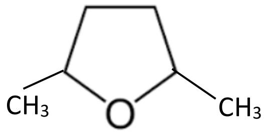
A. I>II>III
B. II>I>III
C. III>II>I
D. I>III>II

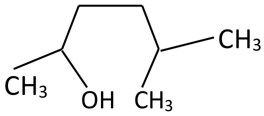
35. 

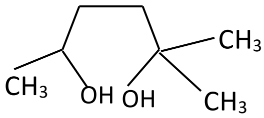

$\xrightarrow[\text{(II) } H_3O^+]{\text{(I) excess CH3MgBr/ether}}$

The product would be

A. 


B. 

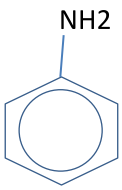

C. 


D. 

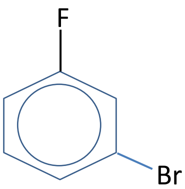

36. The correct decreasing order of acid strength of the following compounds is

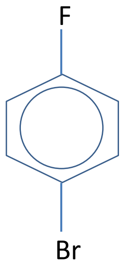
| | | |
|---|---|---|
| $CH_3COOH$ | $CH_3OCH_2COOH$ | $CF_3COOH$ |
| I | II | III |

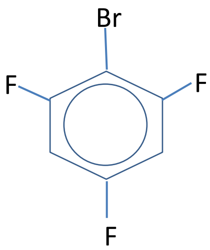
A. I>II>III
B. III>II>I
C. III>I>II
D. II>III>I

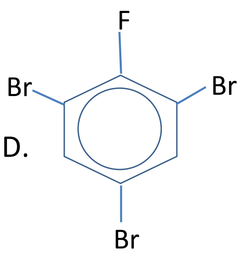
37. 


$\xrightarrow[\text{(ii)} HNO_2 \text{ (0-5}^0\text{C)} \text{ (iii)} HBF_4, \Delta]{\text{(i) } Br_2/H_2O}$ A, The major product(A) is

A. 

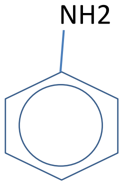

B. 

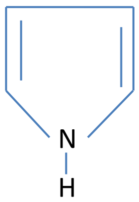

C. 

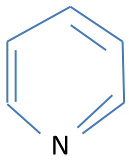

D. 

38. Which of the following compounds react with aldehyde to form Schiff's base?
   A. $1^0$ amines
   B. $2^0$ amines
   C. $3^0$ amines
   D. Amides

39. The correct decreasing order of basicity of the following compounds is

I. 

II. 

III. 

   A. II>I>III
   B. III>I>II
   C. II>IIII>I
   D. I>II>III

40. The reason for double helical structure of DNA is operation of:
   A. Van der waals forces
   B. Hydrogen bonding
   C. Dipole-dipole moment
   D. Electrostatic attractions

41. Glucose reacts with acetic anhydride to form
   A. Hexa acetate
   B. Hepta acetate
   C. Tri acetate
   D. Pena acetate

42. Which of the following compounds react with ferric ions to produce blood red colouration in the Lassaigne's test for detection of elements in an organic compound?
   A. NaCN
   B. $Na_2S$
   C. NaSCN
   D. NaCNO

43. Before testing for halogens, the Lassaigne's extract is boiled with
   A. Conc. HCl
   B. NaOH
   C. Conc. $HNO_3$
   D. Any of these.

44. A salt is heated first with dil. $H_2SO_4$ and then with conc. $H_2SO_4$. No reaction takes place. It may be

A. Nitrate
B. Sulphide
C. Sulphite
D. Sulphate

45. Which of the following amines, on reaction with benzenesulphonyl chloride, will give a sulphonamide that is insoluble in alkali?

A. Ethylamine
B. Ethylmethylamine
C. Trimethylamine
D. Aniline

***ANSWER KEY***

| QUESTION | ANSWER | QUESTION | ANSWER |
|---|---|---|---|
| 1 | D | 24 | B |
| 2 | A | 25 | A |
| 3 | C | 26 | A |
| 4 | A | 27 | D |
| 5 | B | 28 | A |
| 6 | C | 29 | A |
| 7 | D | 30 | D |
| 8 | A | 31 | B |
| 9 | C | 32 | B |
| 10 | D | 33 | A |
| 11 | A | 34 | C |
| 12 | A | 35 | D |
| 13 | B | 36 | B |
| 14 | B | 37 | D |
| 15 | D | 38 | A |
| 16 | C | 39 | B |
| 17 | A | 40 | B |
| 18 | B | 41 | D |
| 19 | D | 42 | C |
| 20 | C | 43 | C |
| 21 | D | 44 | D |
| 22 | A | 45 | B |
| 23 | B | | |